MIL-D-28755(EC)
AMENDMENT 3
8 May 1974
SUPERSEDING 1/
AMENDMENT 2
20 September 1973

MILITARY SPECIFICATION

DECODER GROUP, AN/UPA-60(V) IFF MK XII (SIF), WITH ACTIVE READOUT AND ASSOCIATED FUNCTIONS

This amendment forms a part of Military Specification MIL-D-28755(EC), 16 November 1970.

Page 1

2.1, SPECIFICATIONS, MILITARY: Add the following:

"MIL-E-5400 - Electronic Equipment, Airborne, General Specification for"

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2.1, STANDARDS, MILITARY: Add the following:

"MIL-STD-704 - Electric Power, Aircraft, Characteristics and Utilization of"

"MIL-STD-883 - Test Methods and Procedures for Microelectronics"

Page 3

3.2, line 2: After "specified" add the following:

"and in accordance with the provisions of MIL-E-5400 (Class 1A), to the extent specified herein. In the event of conflict the provisions of MIL-E-5400 shall govern over the provisions of MIL-E-16400."

3.2.1: Delete lines 10 and 13 beginning "(2) Emergency" and "(4) PPI," respectively.

3.2.1.1, lines 1 and 2: Delete "interrogators" through "AN/UPS-1( )." and substitute "Interrogator set AN/APX-76 and its associated coder/ synchronizer unit."

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3.3.5: Delete and substitute the following:

"3.3.5 Maintainability.

"3.3.5.1 Quantitative maintainability requirement. The equipment shall have a specified equipment repair time (ERT) of 15 minutes with repair

1/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, AND SO FORTH) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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accomplished by the replacement of modules, chassis-mounted electronic, electro-mechanical, and mechanical parts. The equipment repair time (ERT) includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks.

3.3.5.2 Qualitative maintainability requirement. The equipment shall be" constructed to provide ease of maintenance, accessibility and replacement of all modules and parts.

3.3.5.3 Maintainability program plan. The contractor shall prepare a maintainability program plan in accordance with the requirements of MIL-STD-470. The maintainability program plan shall describe the tasks to be performed and the procedures for conducting and controlling the maintainability program. The plan shall address each paragraph of MIL-STD-470 in sequence. The plan shall define and identify the interface between the maintainability and integrated logistic support requirements (See 6.3).

3.3.5.4 Maintainability prediction and analysis. A maintainability prediction shall be performed using Part A (to determine the $MTTR_G$) and B (to determine the mean preventive maintenance) Procedure II of MIL-HDBK-472. The contractor shall document and justify all assumptions and the applicability of all data used in the prediction. The prediction shall include all mechanical, electro-mechanical, and electronic parts. When repair is accomplished primarily by module replacement, the prediction shall be prepared in two parts. The prediction shall be made for replacement of modules and chassis-mounted components and parts at the organizational level of maintenance. The results of the prediction shall be compared to the required ERT. Where the predicted values do not achieve the requirements the contractor shall accomplish such changes in design as are necessary to improve the predicted values sufficiently to achieve the requirements (See 6.3).

3.3.5.5 Maintainability reports. Reports in accordance with MIL-STD-470 that provide a complete accounting of progress on each element of the plan, results achieved, procedures used, and status of actions to resolve problems and correct deficiencies shall be prepared (See 6.3).

3.3.6: Delete

Add the following new paragraph:

"3.4.1.2 Plastic Integrated Circuits (IC's) shall not be used and microcircuits employed shall be tested and inspected in accordance with MIL-STD-883, Method 5004, for Class B devices".

Add the following new paragraph:

"3.4.3.2 In addition to meeting the environmental conditions as specified herein under MIL-E-16400, the equipment shall be capable of operation under the temperature-altitude (Class IA), vibration (curves IA and IIA), shock, humidity, and electromagnetic interference environmental conditions as specified herein under MIL-E-5400. The electromagnetic interference characteristics test results shall be obtained in accordance with MIL-E-5400."

3.4.4, line 3: Delete "MIL-E-16400" and substitute "MIL-STD-704, Category B."

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3.4.5.1: Delete lines 1 and 2 and substitute the following:

"Each equipment is not required to be furnished with a set of 10 foot cables (complete and wired with connectors). However, the set of cables furnished with the GFE AN/UPA-60(V2) contractor furnished Interrogator and PPI type display (which includes contractor furnished area gate, such as, range/azimuth PPI modification feature) shall be employed during first article testing to aid in verification of equipment interconnection and operation with AN/APX-76 for the following:"

3.4.5.1.1: Delete the second sentence.

3.4.8.1, line 3: Delete "class 1C of MIL-STD-461" and substitute "MIL-E-5400."

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3.4.9, line 1: Delete "inclination, lab and road."

3.4.9.1: Delete and substitute the following:

"3.4.9.1 Shock and vibration. The equipment shall be capable of withstanding the shock and vibration (curves IA and IIA) requirements of MIL-E-5400 (see 4.5.12)."

3.4.9.2: Delete.

3.4.9.3: Delete.

3.4.10: Line 2, delete "9 (width)" and substitute "10-3/8 (width)." Line 6, delete "and the new logic card requirement (see 3.6.7)." and substitute "."

3.4.11: Delete the first sentence.

3.4.12, line 1: Delete "In the transport condition,"

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3.4.20(a)(1): Delete "M^3IX" and substitute "MIX"

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3.4.22: Add the following at the end of the paragraph:

"The first article inspection (see 4.3) and production control inspection (see 4.4.2) shall demonstrate interchangeability on

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a two-way, forward and backward basis between the GFE and the production equipment. The interchangeability shall be down to the levels as follows:

(a) each plug-in printed circuit card as an entity, maintaining input and output values and their tolerances to achieve performance as required herein.

(b) each subassembly, assembly, and each plug-in module, such as, power supply module and active readout unit, as an entity, maintaining inputs, outputs, and tolerances to achieve performance as required herein."

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3.4.25: Delete the last sentence.

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3.4.30, line 5: Delete "attitude" and substitute "altitude"

3.4.32: Delete subparagraph (a) and substitute the following:

"(a) Each equipment shall be provided with test fittings as necessary to permit maintenance, calibration, and repair with test equipment selected for the purpose. General purpose electronic test equipment shall be selected from MIL-STD-1364(Navy) Standard General Purpose Electronic Test Equipment. Requests for approval for other than the equipment listed in MIL-STD-1364 shall be in accordance with MIL-STD-1345, Figures 1 and 7. Contractor shall, as soon as practical, submit a recommended list of test equipment proposed for use at the various echelons, or levels of maintenance. Unless otherwise specified in the contract or order, the items of test equipment shall not be furnished."

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3.6.4: Delete the first two sentences and substitute the following:

"An automatic emergency alarm function shall be supplied with features as detailed herein, however, an enclosed speaker suitable for mounting remotely is not required. The GFE Emergency Alarm-Monitor BZ-174/UPA-60(V) shall be employed to aid in performance verification and use of the appropriate lights and tones as detailed herein."

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3.6.6.1.1, line 3: Add the following between "unit" and "shall":

", employing Light Emitting Diode (LED) devices,"

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3.6.6.1.1, line 6: Add the following between "cabling." and "Logic":

"Therefore, the only circuitry allowed in the active readout unit is the readout device itself, the current drivers, BCD conversion, drive for LED display, and the LED display elements. In addition, the complete active readout unit shall be functionally, mechanically and electrically interchangeable with the complete active readout unit of the GFE AN/UPA-60(V2)."

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3.6.6.1.5, line 4: Delete "required" and substitute "described"

3.6.6.1.8: Delete

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3.6.6.1.10 through 3.6.7.7.7: Delete

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4.2.1: Add the following at the end of the paragraph:

"Performance parameters to be measured during the individual tests, or tests combined as feasible, shall be those equipment parameters submitted concurrently and integral with the test procedures, and which are approved as part of the procedures. The foregoing also applies to the period or time intervals during the tests when the parameters are to be measured."

4.2, lines 7 and 8: Delete subparagraph beginning "(d) Inspection" and substitute the following:

"(d) Maintainability Demonstration (see 4.6)

(e) Inspection of preparation for delivery"

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4.2.2 through 4.2.2.3: Delete and substitute the following:

"4.2.2 Reliability assurance tests. Reliability assurance tests shall be conducted using MIL-STD-781. Tests as required by both the qualification phase and the production acceptance (sampling) phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.2.2.1 Qualification (demonstration) phase. Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled Qualification Phase of Production Reliability Tests. The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the qualification phase, test level E shall be used. The accept-reject criteria for test plan IV shall be used.

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4.2.2.2 Reliability production acceptance (sampling) phase tests. The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled Production Acceptance (Sampling) Phase of Production Reliability Tests. Test level E shall be used.

4.2.2.2.1 All equipment test. Each equipment produced except those submitted for the reliability qualification test, shall be tested for 40 hours. Prior to the 40 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of test plan II of MIL-STD-781. (Extend the line as necessary to accomodate the date.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, and so forth., so that the entire compilation will show other than a reject decision.

4.2.2.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, and so forth, shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests."

4.2.2.9: Delete

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Table I: Line 11, delete "Electromagnetic Interference Emission and Susceptibility and Self-compatibility" and substitute "Electromagnetic Interference Characteristics".

Line, 15 delete "and Inclination 4.5.13"

Line, 16 delete "Sand and Dust Test 4.5.14"

Line, 17 delete "Salt Spray Test 4.5.15"

Line, 19 delete "Road Test 4.5.17"

Line, 22 delete "Lab Package Test 4.5.20"

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4.4.2: Delete the second and third sentences and substitute the following: "The samples for production control inspection shall be randomly selected and shall be one unit for each 15 production units".

Table II: Line 7, delete "Electromagnetic interference emission and susceptibility and self-compatibility" and substitute "Electromagnetic Interference Characteristics".

Add the following new paragraph:

"4.4.2.4 Electromagnetic interference tests (EMI) - EMI tests for production control inspection shall be performed on one unit of the first eight production units and on one unit of the next to the last eight production units (or fraction thereof as practical)".

Table III: Line 3, delete "and Inclination 4.5.13". Line 4, delete "Salt Spray 4.5.15".

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4.4.3.1(2): Delete "first 25" and substitute "first 50".

4.5.4, line 2: Delete "MIL-E-16400" and substitute "MIL-STD-704, Category B."
4.5.6(a): Delete

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4.5.7.2: Delete and substitute the following:

"4.5.7.2 Temperature-altitude. Temperature-altitude tests shall be performed per MIL-STD-810, Method 504 to demonstrate compliance with 3.4.3(a) and 3.4.3.2. Verification shall be made during the test, of compliance with 3.4.4.1 regarding line supply surges, line regulation, and 50 percent overload factor."

4.5.8 and 4.5.8.1: Delete and substitute the following:

"4.5.8 Humidity. Humidity tests shall be performed per MIL-STD-810, Method 507, Procedure I."

4.5.9, line 2: Add "in accordance with MIL-STD-704, Category B." between "measured" and "to".

4.5.10: Delete and substitute the following:

"4.5.10 Electromagnetic interferenced characteristics. Electromagnetic interference characteristics tests shall be performed in accordance with MIL-E-5400 to demonstrate compliance with 3.4.8.1."

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4.5.12: Delete and substitute the following:

"4.5.12 Shock and vibration tests.

"4.5.12.1 Shock tests shall be performed in accordance with MIL-E-5400 and MIL-STD-810, Method 516, Procedures I and III, utilize Figure 516-2. Normal operation shall be ascertained prior to and after completing tests.

"4.5.12.2 Vibration tests shall be performed in accordance with MIL-E-5400, Curves IA and IIA. The dimensional and weight requirements of the end item, as specified herein, are less the mount. The contractor shall furnish the mount and mount configuration for the first article samples only.

4.5.13: Delete.

4.5.14: Delete.

4.5.15: Delete.

4.5.16, line 4: Delete "3.4.22.2." and substitute "3.4.22."

4.5.17: Delete

Add the following new paragraph:

"4.5.18.3 The performance tests of 4.5.18 shall include specifically emphasized demonstration of the following equipment functions:

(a) the active readout display disciplines as detailed in 3.6.6.1.2(a)(1)(2) and (3).
(b) selected altitude layer operation as detailed in 3.6.5, including both single and dual mode challenges and both passively and actively decoded."

4.5.19, line 1: Delete "in the transport condition"

4.5.20, line 1: Delete

Add the following new paragraphs:

"4.6 Maintainability demonstration. The maintainability requirements shall be demonstrated in accordance with Method 4 of MIL-STD-471. The maintainability demonstration shall be successfully completed prior to the delivery of any equipment. The maintainability demonstration test shall be performed and completed by the contractor at the contractors facility. The demonstration shall be limited to Phase 2 (demonstration) of MIL-STD-471.

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"4.6.1 Maintainability demonstration test plan. The contractor shall prepare a detailed maintainability test plan in accordance with MIL-STD-471. The maintainability test at the organizational level shall be performed by a technician with experience equivalent to that of a third class Petty Officer. The equipment maintainability requirements shall be demonstrated by replacement of modules and chassis-mounted electronic, electrical and electromechanical components and parts at the organizational level. A list of 50 faults shall be prepared for the equipment maintainability demonstration in accordance with Appendix A of MIL-STD-471. The command or agency concerned or its representative shall use this list as a guide to select a sample of 20 faults to be used in this part of the maintainability demonstration. For fault simulation, a number shall be assigned to each component and part within a module or chassis-mounted component and part. Fault simulation for modules shall be accomplished by introduction of faulty parts corresponding to the number selected.

"4.6.2 Technicians. The command or agency concerned reserves the right to provide and select the technicians to perform either or both the equipment and module maintainability demonstration. The contractor will provide adequate training to qualify the military technician(s) to perform the maintainability demonstration.

"4.6.3 Technical documentation. Technical documentation to be used shall be limited to the equipment technical manual. If the technical manual is not available, a draft shall be substituted upon approval of the command or agency concerned.

"4.6.4 Rejection. If a reject decision is reached, the command or agency concerned shall be immediately notified. The contractor shall, at no additional cost to the Government:

(a) Develop a plan for design or correction of all deficiencies.
(b) Upon approval of the plan perform corrective action and repeat the demonstration until an accept decision is reached.

"4.6.5 Maintainability demonstration report. A maintainability demonstration report shall be prepared in accordance with MIL-STD-471. The report shall include an analysis of the statistical distribution of the repair time."

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6.3: Delete (f) and substitute the following:

"(f) Maintainability reports (see 3.3.5.5)

6.3: Add the following:

(1) Maintainability demonstration report (see 4.6.5)

Preparing activity:
NAVY-EC

(Project 5895-N239)

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